Ashutosh Jarag

AI Engineer | ML Engineer | LLM Engineer | Python · LLMs · RAG · LangChain · FastAPI
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SUMMARY

AI & Machine Learning Engineer with 14 months of internship experience building **Generative AI, LLM, RAG, NLP** and **machine learning** applications using **Python, LangChain, LangGraph, Hugging Face, FastAPI** and **vector databases**. Built **RAG** systems, **text-to-SQL** chatbots and an **LLM document extraction** pipeline, with hands-on experience in **semantic search, embeddings, prompt engineering** and **ML inference APIs**. Also experienced in **ETL pipelines, PostgreSQL, Docker** and **AWS**.

EXPERIENCE

AI/ML Intern Jan 2025 – Nov 2025
Compserv Consultant Pvt Ltd *Kolhapur, Maharashtra*

- Built an **LLM-powered document extraction pipeline** for handwritten bills (extract, clean, validate, review in a form, store in **PostgreSQL**), reducing manual effort by 70%.
- Built a **RAG** application for the company website using **LangChain, embeddings** and **vector databases** for **semantic search** and context-aware answers.
- Developed a **text-to-SQL** chatbot for the support team using **LLMs, SQL, LangChain** and **LangSmith**.
- Developed **FastAPI** backend services and a **secure file-storage and tracking system** with encryption and audit trails on **PostgreSQL**.
- Analyzed client datasets with **SQL** and **Python** (**EDA**, data cleaning) and contributed to **R&D** on **LLM evaluation**, vector database and framework selection, and system architecture.

Data Science Intern Jun 2024 – Aug 2024
Tvarit Solutions Pvt Ltd *Pune, Maharashtra*

- Developed **automated Python pipelines** to extract data from **AWS cloud storage, PostgreSQL databases** and flat files, apply preprocessing and generate business reports.
- Cleaned and analyzed **industrial data** (**EDA**, statistical analysis) to support senior engineers and data scientists in model and project development.
- Refactored procedural scripts into **modular Python scripts**, improving code readability and maintainability.

TECHNICAL SKILLS

Programming Languages: Python, SQL, Bash/Shell Scripting
Generative AI & LLMs: LLMs, Generative AI, RAG, Prompt Engineering, LangChain, LangGraph, LangSmith, Gemini, Embeddings, Semantic Search, MCP, Guardrails
Vector Databases: Qdrant, ChromaDB, Pinecone, FAISS
Machine Learning: Scikit-learn, XGBoost, Supervised & Unsupervised Learning, Classification, Regression, Clustering, Feature Engineering, Model Evaluation, Hyperparameter Tuning
Deep Learning & NLP: PyTorch, Neural Networks, NLP, Transformers, Hugging Face, Sentence Transformers, TF-IDF
Backend & Cloud: FastAPI, REST APIs, Pydantic, Docker, Git, GitHub Actions (CI/CD), Linux, AWS
Databases & Data: PostgreSQL, MySQL, MongoDB, PySpark, ETL Pipelines, EDA, Tableau, Power BI

EDUCATION

Centre for Development of Advanced Computing (CDAC) Pune, Maharashtra
PG-Diploma in Big Data Analytics *Feb 2026 – Aug 2026*

Shivaji University Kolhapur, Maharashtra
B.Tech in Computer Science (Data Science) - 8.2 CGPA *Sep 2022 – May 2025*

Dr. D. Y. Patil Polytechnic Kolhapur, Maharashtra
Diploma in Computer Science - 84.23 % *Sep 2019 – May 2022*

PROJECTS

AI-Powered Research Paper Intelligence System | *Python, NLP, ML, LLM, RAG, ChromaDB, FastAPI* *GitHub*

- Built an AI research platform processing **3.1M+ arXiv papers** using **NLP, machine learning, semantic search** and **RAG** for large-scale research discovery.
- Implemented paper **classification (TF-IDF + SVM)**, publication prediction, **semantic recommendations**, summarization and document-based **question answering** with **Gemini** and RAG.
- Integrated **ML inference, ChromaDB** vector retrieval and **Gemini** workflows through a modular **FastAPI** backend with **Pydantic** validation and reusable service layers.

AI Math Agent | *Python, FastAPI, Gemini, Qdrant, RAG, MCP, Docker, PostgreSQL* *GitHub*

- Developed an AI math assistant using **Gemini, Qdrant** and **FastAPI** that generates step-by-step solutions through a **RAG** pipeline with **semantic retrieval** over a knowledge base.
- Added **MCP**-based web-search fallback, **feedback-based retraining** and **guardrails** to improve response quality and relevance.

AutoLog: Smart Vehicle Logging System | *Python, FastAPI, React, PostgreSQL* *GitHub*

- Built an **AI-based vehicle logging system** that automates **license plate detection**, vehicle tracking and parking management with a real-time monitoring dashboard on **FastAPI, React** and **PostgreSQL**.

CERTIFICATIONS & LEADERSHIP

Certifications: Cisco Data Analytics, Cisco Introduction to Data Science, Power BI, Advanced SQL
Achievements: Winner, AI/ML TechSphere college tournament; data team lead and backend developer, Mahila Bachat Gat platform (KIC)